

Line Number	Error	Error Type	Error Correction
12	Uninitialized pointer ptr1 dereferenced	Runtime Error	Initialize ptr1 before use
16	NULL pointer ptr2 dereferenced	Runtime Error	Allocate memory before dereferencing
25	Pointer ptr3 increment may access invalid memory	Logical/Runtime Error	Avoid incrementing beyond valid variable
30	Out-of-bounds access *(ptr4 + 10)	Runtime Error	Access within array bounds
37	Use-after-free ptr5	Runtime Error	Do not access memory after free
44	Memory leak ptr6	Memory Leak	Free memory before setting pointer to NULL
49	Insufficient allocation for ptr7	Runtime Error	Use malloc(5 * sizeof(int))
51	Loop writes beyond allocated memory ptr7	Runtime Error	Limit loop to allocated size
60	Out-of-bounds ptr8[3]	Runtime Error	Allocate enough memory or limit index
69	Double free ptr9	Runtime Error	Free memory only once
75	Uninitialized ptr10 dereferenced	Runtime Error	Initialize allocated memory before use
82	Out-of-bounds loop numbers array	Runtime Error	Use i < 5
96	Incorrect double pointer assignment	Type Error	Use int *p; pptr = &p;
103	Missing string.h for strcpy	Compilation Error	Include <string.h>
104	Buffer overflow in strcpy	Runtime Error	Allocate sufficient memory
111	Pointer increment ptr11 causes invalid access	Runtime Error	Do not increment beyond allocated memory
115	NULL pointer dereference ptr12	Runtime Error	Check pointer before dereferencing
124	Out-of-bounds ptr13[100]	Runtime Error	Use index < 100
131	Use-after-free ptr14	Runtime Error	Do not use pointer after free
138	Invalid matrix pointer arithmetic	Runtime Error	Use correct indexing
145	Memory leak ptr15	Memory Leak	Free previously allocated memory before reassigning
152	Swap function incorrect logic	Logical Error	Swap values using *a and *b
160	printArray out-of-bounds loop	Runtime Error	Use i < size